

DBSE & DBD Practical Week-3

```
p_user
'mongosh' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\vishnuvardhan>mongodb+srv://cluster0.ehdoarm.mongodb.net/
'mongodb+srv:' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\vishnuvardhan>mongosh "mongodb+srv://cluster0.ehdoarm.mongodb.net/" --username vadakattu_vishnuvardhan_kumar_db_user
'mongosh' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\vishnuvardhan>C:\Users\vishnuvardhan\Downloads\mongosh-2.10.0-win32-x64\mongosh-2.10.0-win32-x64\bin
C:\Users\vishnuvardhan\Downloads\mongosh-2.10.0-win32-x64\mongosh-2.10.0-win32-x64\bin' is not recognized as an internal
external command,
operable program or batch file.

C:\Users\vishnuvardhan>cd C:\Users\vishnuvardhan\Downloads\mongosh-2.10.0-win32-x64\mongosh-2.10.0-win32-x64\bin

C:\Users\vishnuvardhan\Downloads\mongosh-2.10.0-win32-x64\mongosh-2.10.0-win32-x64\bin>mongosh "mongodb+srv://cluster0.eh
mongodb.net/" --username vadakattu_vishnuvardhan_kumar_db_user
Enter password: *****
Current Mongosh Log ID: 6a89300b3bf149dce656a078
Connecting to:      mongodb+srv://<credentials>@cluster0.ehdoarm.mongodb.net/?appName=mongosh+2.10.0
Using MongoDB:      8.0.29
Using Mongosh:       2.10.0

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

Atlas atlas-q6fn0f-shard-0 [primary] test> |
```

```
publishedYear: 2021,
formats: ["Digital", "Paperback"],
specs: {
  pages: 210,
  fileSize_MB: 2.1,
  drm: false
},
avgRating: 4.2,
reviews: [
  {
    member_id: "M2001",
    rating: 4,
    comment: "Solid Digital read, clean layout.",
    date: ISODate("2022-05-10")
  },
  {
    member_id: "M2002",
    rating: 5,
    comment: "Best programming book this year.",
    date: ISODate("2022-06-01")
  }
]
},
{
  title: "Whispers of Yesterday",
  author: "S. Kapoor",
  genre: "Fiction",
  publishedYear: 2019,
  formats: ["Hardcover", "Paperback"],
  specs: {
    pages: 410,
    binding: "Sewn"
  },
  avgRating: 3.8,
  reviews: [
    {
      member_id: "M3001",
      rating: 3,
      comment: "Slow start but strong ending.",
      date: ISODate("2020-01-15")
    },
    {
      member_id: "M3002",
      rating: 4,
      comment: "Beautifully written prose.",
      date: ISODate("2020-02-11")
    }
  ]
}
```

```

    member_id: "M3002",
    rating: 4,
    comment: "Beautifully written prose.",
    date: ISODate("2020-02-11")
  }
]
},
{
  title: "Quantum Horizons",
  author: "R. Menon",
  genre: "Science",
  publishedYear: 2023,
  formats: ["Digital", "Audiobook"],
  specs: {
    fileSize_MB: 5.6,
    drm: true,
    narrator: "P. Iyer",
    runtime_min: 540
  },
  avgRating: 4.9,
  reviews: [
    {
      member_id: "M4001",
      rating: 5,
      comment: "The Digital format made annotations so easy.",
      date: ISODate("2023-08-01")
    },
    {
      member_id: "M4002",
      rating: 5,
      comment: "Mind-bending and well narrated.",
      date: ISODate("2023-09-14")
    }
  ]
}
}
])
}
}
acknowledged: true,
insertedIds: {
  '0': ObjectId('6a8930da3bf149dce656a079'),
  '1': ObjectId('6a8930da3bf149dce656a07a'),
  '2': ObjectId('6a8930da3bf149dce656a07b'),
  '3': ObjectId('6a8930da3bf149dce656a07c')
}
}
}
Atlas atlas-g6fn0f-shard-0 [primary] BookFlow> db.book_metadata.countDocuments

```

```
gitVersion: '559d67c651f7393e62757234a0b25cb7a8622148'
},
serverParameters: {
  internalQueryFacetBufferSizeBytes: 104857600,
  internalQueryFacetMaxOutputDocSizeBytes: 104857600,
  internalLookupStageIntermediateDocumentMaxSizeBytes: 16793600,
  internalDocumentSourceGroupMaxMemoryBytes: 104857600,
  internalQueryMaxBlockingSortMemoryUsageBytes: 33554432,
  internalQueryProhibitBlockingMergeOnMongoS: 0,
  internalQueryMaxAddToSetBytes: 104857600,
  internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
  internalQueryFrameworkControl: 'trySbeRestricted',
  internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
},
ok: 1,
'$clusterTime': {
  clusterTime: Timestamp({ t: 1787375911, i: 1 }),
  signature: {
    hash: Binary.createFromBase64('4cmJyxRJNM3joq0EftPhq4J32DQ=', 0),
    keyId: Long('7631615457119698946')
  }
},
operationTime: Timestamp({ t: 1787375911, i: 1 })
}
}
}

tlsh atlas-q6fn0f-shard-0 [primary] BookFlow> db.book_metadata.find(
{
  "reviews.comment": {
    $regex: "Digital",
    $options: "i"
  }
}
).explain("executionStats")

explainVersion: '1',
queryPlanner: {
  namespace: 'BookFlow.book_metadata',
  parsedQuery: { 'reviews.comment': { '$regex': 'Digital', '$options': 'i' } },
  indexFilterSet: false,
  queryHash: '9DF7CB16',
  planCacheShapeHash: '9DF7CB16',
  planCacheKey: 'A9183C86',
  optimizationTimeMillis: 0,
  maxIndexedOrSolutionsReached: false,
  maxIndexedAndSolutionsReached: false,
  maxScansToExplodeReached: false,
}
```

```

y()
[
  {
    _id: ObjectId('6a8930da3bf149dce656a079'),
    title: 'The Silent Algorithm',
    author: 'R. Menon',
    genre: 'Technology',
    publishedYear: 2022,
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ],
    specs: {
      pages: 340,
      fileSize_MB: 4.2,
      drm: true,
      narrator: 'K. Rao',
      runtime_min: 610
    },
    avgRating: 4.6,
    reviews: [
      {
        member_id: 'M1001',
        rating: 5,
        comment: 'Excellent deep dive into digital systems design.',
        date: ISODate('2023-01-15T00:00:00.000Z')
      },
      {
        member_id: 'M1042',
        rating: 4,
        comment: 'Good but the Digital edition had formatting issues.',
        date: ISODate('2023-02-20T00:00:00.000Z')
      }
    ]
  },
  {
    _id: ObjectId('6a8930da3bf149dce656a07a'),
    title: 'Ocean of Code',
    author: 'A. Fernandes',
    genre: 'Technology',
    publishedYear: 2021,
    formats: [ 'Digital', 'Paperback' ],
    specs: { pages: 210, fileSize_MB: 2.1, drm: false },
    avgRating: 4.2,
    reviews: [
      {
        member_id: 'M2001',
        rating: 4,
        comment: 'Solid Digital read, clean layout.',

```

```

60F3B51',
  command: {
    find: 'book_metadata',
    filter: { 'reviews.comment': { '$regex': 'Digital', '$options': 'i' } },
    '$db': 'BookFlow'
  },
  serverInfo: {
    host: 'ac-en6usc8-shard-00-02.ehdoarm.mongodb.net',
    port: 27017,
    version: '8.0.29',
    gitVersion: '559d67c651f7393e62757234a0b25cb7a8622148'
  },
  serverParameters: {
    internalQueryFacetBufferSizeBytes: 104857600,
    internalQueryFacetMaxOutputDocSizeBytes: 104857600,
    internalLookupStageIntermediateDocumentMaxSizeBytes: 16793600,
    internalDocumentSourceGroupMaxMemoryBytes: 104857600,
    internalQueryMaxBlockingSortMemoryUsageBytes: 33554432,
    internalQueryProhibitBlockingMergeOnMongoS: 0,
    internalQueryMaxAddToSetBytes: 104857600,
    internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
    internalQueryFrameworkControl: 'trySbeRestricted',
    internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
  },
  ok: 1,
  '$clusterTime': {
    clusterTime: Timestamp({ t: 1787375921, i: 2 }),
    signature: {
      hash: Binary.createFromBase64('mhV+TQ8CZ17cZmaBbQcW9iJ70D0=', 0),
      keyId: Long('7631615457119698946')
    }
  },
  operationTime: Timestamp({ t: 1787375921, i: 2 })
}
Atlas atlas-q6fn0f-shard-0 [primary] BookFlow> db.book_metadata.createIndex(
{
  |   formats: 1
  | })
formats_1
Atlas atlas-q6fn0f-shard-0 [primary] BookFlow> db.book_metadata.createIndex(
{
  |   avgRating: -1
  | })
avgRating_-1
Atlas atlas-q6fn0f-shard-0 [primary] BookFlow> db.book_metadata.find(
{
  |

```

```

maxIndexedOrSolutionsReached: false,
maxIndexedAndSolutionsReached: false,
maxScansToExplodeReached: false,
prunedSimilarIndexes: false,
winningPlan: {
  isCached: false,
  stage: 'FETCH',
  inputStage: {
    stage: 'IXSCAN',
    keyPattern: { avgRating: -1 },
    indexName: 'avgRating_-1',
    isMultiKey: false,
    multiKeyPaths: { avgRating: [] },
    isUnique: false,
    isSparse: false,
    isPartial: false,
    indexVersion: 2,
    direction: 'forward',
    indexBounds: { avgRating: [ '[inf.0, 4)' ] }
  }
},
rejectedPlans: []
},
executionStats: {
  executionSuccess: true,
  nReturned: 3,
  executionTimeMillis: 1,
  totalKeysExamined: 3,
  totalDocsExamined: 3,
  executionStages: {
    isCached: false,
    stage: 'FETCH',
    nReturned: 3,
    executionTimeMillisEstimate: 0,
    works: 4,
    advanced: 3,
    needTime: 0,
    needYield: 0,
    saveState: 0,
    restoreState: 0,
    isEOF: 1,
    docsExamined: 3,
    alreadyHasObj: 0,
    inputStage: {
      stage: 'IXSCAN',
      nReturned: 3,

```

```

    }
  },
  queryShapeHash: 'BBD6A60E9A02C5F80F59ACE33E48FCA9EEA899B60BF36D4840BE3297E
A47B9E4',
  command: {
    find: 'book_metadata',
    filter: { avgRating: { '$gt': 4 } },
    '$db': 'BookFlow'
  },
  serverInfo: {
    host: 'ac-en6usc8-shard-00-02.ehdoarm.mongodb.net',
    port: 27017,
    version: '8.0.29',
    gitVersion: '559d67c651f7393e62757234a0b25cb7a8622148'
  },
  serverParameters: {
    internalQueryFacetBufferSizeBytes: 104857600,
    internalQueryFacetMaxOutputDocSizeBytes: 104857600,
    internalLookupStageIntermediateDocumentMaxSizeBytes: 16793600,
    internalDocumentSourceGroupMaxMemoryBytes: 104857600,
    internalQueryMaxBlockingSortMemoryUsageBytes: 33554432,
    internalQueryProhibitBlockingMergeOnMongoS: 0,
    internalQueryMaxAddToSetBytes: 104857600,
    internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
    internalQueryFrameworkControl: 'try$restricted',
    internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
  },
  ok: 1,
  '$clusterTime': {
    clusterTime: Timestamp({ t: 1787375938, i: 4 }),
    signature: {
      hash: Binary.createFromBase64('2CY74A+ldvaR76/+T1aRqByDL44=', 0),
      keyId: Long('7631615457119698946')
    }
  },
  operationTime: Timestamp({ t: 1787375938, i: 4 })
}
Atlas atlas-q6fn0f-shard-0 [primary] BookFlow> db.book_metadata.countDocumen
ts()
4
Atlas atlas-q6fn0f-shard-0 [primary] BookFlow> |

```

CLUSTERS (1)

Search clusters

Cluster0

BookFlow

book_metadata

admin

cooldb

local

sample_mflix

test

Type a query: { field: 'value' } or [Generate query](#)

Explain

Reset

Find

Options

ADD DATA

BULK

EXPORT CODE

25

1 - 4 of 4

_id: ObjectId('6a8930da3bf149dce656a07b')

title: "Whispers of Yesterday"

author: "S. Kapoor"

genre: "Fiction"

publishedYear: 2019

formats: Array (2)

specs: Object (2)

avgRating: 3.8

reviews: Array (2)

_id: ObjectId('6a8930da3bf149dce656a07c')

title: "Quantum Horizons"

author: "R. Menon"

genre: "Science"

publishedYear: 2023

formats: Array (2)

specs: Object (4)

avgRating: 4.9

reviews: Array (2)

ORGANIZATION
Bharath kumar's Org ...

PROJECT
Project 0

?

Data Explorer

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

BookFlow

book_metadata

admin

cooldb

local

sample_mflix

test

book_metadata

Cluster0 > BookFlow > book_metadata

Documents 4

Aggregations

Schema

Indexes 4

Validation

Type a query: { field: 'value' } or [Generate query](#)

Explain

Reset

Find

Options

ADD DATA

BULK

EXPORT CODE

25

1 - 4 of 4

_id: ObjectId('6a8930da3bf149dce656a079')

title: "The Silent Algorithm"

author: "R. Menon"

genre: "Technology"

publishedYear: 2022

formats: Array (3)

specs: Object (5)

avgRating: 4.6

reviews: Array (2)

_id: ObjectId('6a8930da3bf149dce656a07a')

title: "Ocean of Code"

author: "A. Fernandes"

genre: "Technology"

publishedYear: 2021

formats: Array (2)

specs: Object (3)

avgRating: 4.2

reviews: Array (2)

Alert System Status: All Good